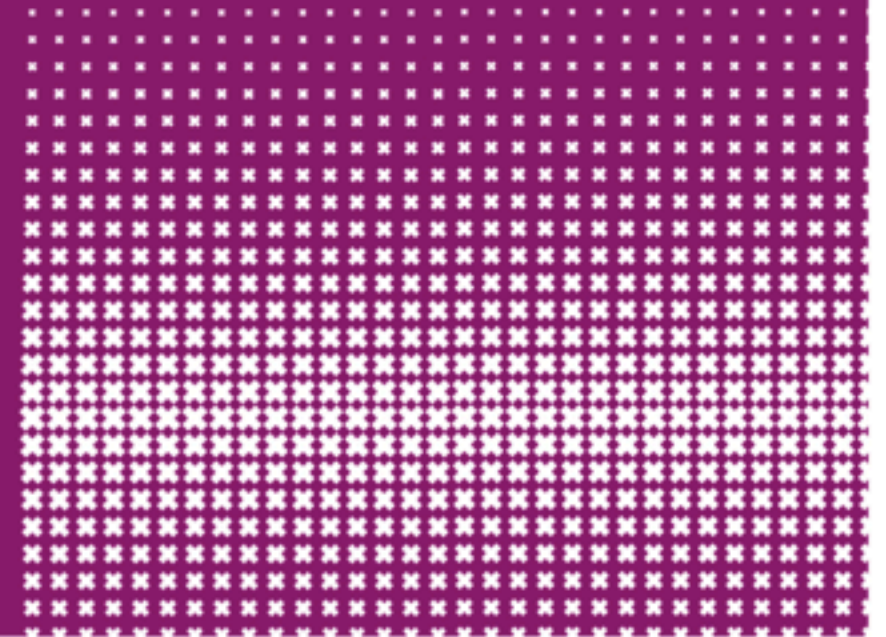




NER and Wikification for DutchSemCor

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Goal

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- Create semantic annotations
 - Useful as features in WSD

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- Two-step approach
 - Identify named entities
 - Link entities to Wikipedia pages

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- Two-step approach
 - Identify named entities
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- Work based on WSDM 2012 paper
 - “Adding Semantics to Microblog Posts”

Menu

- Linking tweets to Wikipedia
- Wikifying Sonar
- Evaluation
- Conclusion

Linking tweets to Wikipedia



Linking tweets to Wikipedia



- Find out what people are saying about entities
 - products, events, brands, people, organizations, etc.
 - entity ~ Wikipedia article

Linking tweets to Wikipedia



- Find out what people are saying about entities
 - products, events, brands, people, organizations, etc.
 - entity ~ Wikipedia article
- Automatically identify concepts (entities) that are *semantically related* to a tweet and generate links to the corresponding Wikipedia articles
 - mentioned
 - implied
 - inferred
 - ...

Data

- Over 500 manually annotated (= linked) tweets
 - annotations publicly available for follow-up research
 - separate labels for 'erroneous' and 'ambiguous'
 - 2.17 entities per tweet on average
- For example...

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 - Cee Lo Green
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- *I would like to congratulate Qatar on their win to host the World Cup in 2022. This victory is an achievement for all Arab countries*
 - Arab World
 - FIFA World Cup
 - 2018 and 2022 FIFA World Cup bids
 - Qatari 2022 FIFA World Cup bid
 - Qatar

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 - Lexical matching
 - Language modeling
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- Step 1 – Ensuring **recall**: obtain an exhaustive ranked list of candidate concepts
 - Lexical matching
 - Language modeling
 - “Other methods,” including DBpedia Spotlight, M&W, etc.
- Step 2 – Boosting **precision**: use supervised machine learning to determine which of the candidate concepts to keep
 - NB, C4.5, SVM, Random Forests, (Initialized) Gradient Boosted Regression Trees

N-gram features

$LEN(q) = q $	Number of terms in the n-gram q
$IDF_f(q)$	Inverse document frequency of q in representation f , where $f \in \{\text{title, anchor, content}\}$
$WIG(q)$	Weighted information gain using top-5 retrieved concepts
$KEYPHRASENESS(q)$	Probability that q is used as an anchor text in Wikipedia
$LINKPROB(q)$	Probability that q is used as an anchor text in Wikipedia (all occurrences)
$SNIL(q)$	Number of articles whose title equals a sub-n-gram of q
$SNCL(q)$	Number of articles whose title match a sub-n-gram of q

Concept features

$INLINKS(c)$	Number of Wikipedia articles linking to c
$OUTLINKS(c)$	Number of Wikipedia articles linking from c
$GEN(c)$	Function of depth of c in the Wikipedia category hierarchy
$CAT(c)$	Number of categories associated with c
$REDIRECT(c)$	Number of redirect pages linking to c
$WLEN(c)$	Number of terms in the title of c
$CLEN(c)$	Number of characters in the title of c
$WIKSTATS(c)$	Number of times c was visited in 2010
$WIKISTATSWK(c)$	Number of times c was visited in the seven days before the tweet was posted

N-gram + concept features

$TF_f(c, q) = \frac{n_f(q, c)}{ f }$	Relative phrase frequency of q in representation f of c , normalized by length of f , where $f \in \{\text{title, anchor, first sentence, first paragraph, content}\}$
$POS_n(c, q) = pos_n(q)/ c $	Position of n th occurrence of q in c , normalized by length of c
$SPR(c, q)$	Distance between the first and last occurrence of q in c
$TF \cdot IDF(c, q)$	Importance of q for c
$RIDF(c, q)$	Residual IDF (difference between expected and observed IDF)
$\chi^2(c, q)$	χ^2 test of independence between q in c and all concepts
$NCT(c, q)$	Does q contain the title of c ?
$TCN(c, q)$	Does the title of c contain q ?
$TEN(c, q)$	Does the title of c equal q ?
$SCORE(c, q)$	Language modeling score of c with respect to q
$RANK(c, q)$	Retrieval rank of c with respect to q
$COMMONNESS(c, q)$	Probability of c being the target of a link with anchor text q

Tweet features

$TWCT(c, Q)$	Does Q contain the title of c ?
$TCTW(c, Q)$	Does the title of c contain Q ?
$TETW(c, Q)$	Does the title of c equal Q ?
$TAGDEF(q, Q)$	Number of times q appears in the hashtag definition of any hashtag in tweet Q
$URL(q, Q)$	Number of times q appears in a webpage linked to by Q

Results

■ Step 1 – Ensuring **recall**

	P1	R-prec	Recall	MRR	MAP
Spotlight	0.3717	0.2688	0.3068	0.4215	0.2658
M&W	0.3770	0.3033	0.3741	0.4255	0.3127
Tagme	0.5628	0.4643	0.5814	0.6339	0.4894
CMNS	0.6021	0.5271	0.7775	0.7080	0.5853

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$$CMNS(c, q) = \frac{|L_{q,c}|}{\sum_{c'} |L_{q,c'}|},$$

where $L_{q,c}$ is the set of all links with anchor text q and target c .

Results

■ Step 2 – Boosting **precision**

	P1	R-prec	Recall	MRR	MAP
CMNS	0.6021	0.5271	0.7775	0.7080	0.5853
NB	0.4921▼	0.4091▼	0.5292▼	0.5853▼	0.4294▼
C4.5	0.5209▼	0.3542▼	0.3774▼	0.5585▼	0.3528▼
SVM	0.6708▲	0.4670▼	0.4784▼	0.6846	0.4711▼
RF	0.6780▲	0.5739[△]	0.8417▲	0.7676▲	0.6561▲
GBRT	0.6649 [△]	0.5624 [△]	0.8361▲	0.7568▲	0.6421▲
iGBRT	0.6754 [△]	0.5715 [△]	0.8400▲	0.7644▲	0.6511▲

Analysis

Added feature	MAP	Δ
$TEN(c, q) / TWCT(c, Q)$		
$/ REDIRECT(c) / LINKPROB(q)$	0.4964	
+ $IDF_{anchor}(q)$	0.5452	9.83%
+ $KEYPHRASENESS(q)$	0.5523	1.30%
+ $SNIL(q)$	0.5502	-0.38%
+ $IDF_{title}(q)$	0.5547	0.82%
+ $SNCL(q)$	0.5593	0.83%
+ $TF_{paragraph}(c, q)$	0.5674	1.45%
+ $TF_{sentence}(c, q)$	0.5834	2.82%
+ $TCN(c, q)$	0.5866	0.55%
+ $COMMONNESS(c, q)$	0.6216	5.97%
+ $TF_{title}(c, q)$	0.6340	1.99%
+ $URL(q, Q)$	0.6405	1.03%
+ $POS_1(c, q)$	0.6435	0.47%

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 - ...first run Stanford NER

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- Why not NER?
 - Introduce additional source of errors

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- Only correctly identified entities are considered for wikification evaluation
- Overall results
 - NER: P:0.78, R:0.65
 - WIK: P:0.84, R:0.82

Evaluation II

■ Per document type

	precision	recall
Wikipedia	0.89	0.89
Discussion list	0.75	1.00
Web snippet	1.00	0.25
Autocue	0.96	0.96
Newspaper	0.90	0.85
Periodical	0.95	0.95
E-magazine	0.86	0.89
Legal text	0.56	0.47
Subtitle	0.54	0.49

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 - Identifying NEs
 - Linking NEs to Wikipedia
- Linking to Wikipedia in two steps
 - Recall step: candidate pages
 - Precision step: learn which pages to keep
- Evaluation shows good performance
 - High precision and recall on formal text
 - Lower scores for domain-specific or very informal text